

LUIS ALBERTO ORTIZ MEZA

Solutions Architect | Cloud & Applied AI | Tech Lead

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PROFESSIONAL PROFILE

Solutions architect and technical lead with **14+ years** building secure, regulated software on **AWS** for government, electoral, health, legal and education institutions. **M.Eng. in Information Security (IPN)**. Co-founder and CTO of ZOGA, where I lead **Certisep**, ZOGA's SaaS product for electronic academic credentials. **Applied AI**: an electoral OCR product rated **best of 3 solutions evaluated by INE** (97% reliability), multi-agent pipelines with local LLMs **in production**, and an auditable **RAG agent on AWS Bedrock** (public on GitHub).

Solution Architecture · AWS Serverless · Amazon Bedrock · Generative AI · RAG · Multi-agent LLM Systems · Terraform · Python · Node.js · Security by Design · Technical Leadership

PROFESSIONAL EXPERIENCE

ZOGA mobile & IT | Co-founder & CTO

Mexico City · Mar 2012 – Present

Software development company. I lead the technical vision, solution architecture and development teams of up to **5 engineers**.

Certisep — ZOGA's SaaS product (Legal-Tech) · Oct 2018 – Present

- Architected the platform for issuing, digitally signing and validating **electronic degrees and certificates** (SEP XML standard) for private higher-education institutions, with isolated regulatory workflows for certificates (DGAIR) and degrees (DGP).
- Designed and deployed **multi-agent prospecting pipelines** (CrewAI + Ollama, local LLMs on Linux): weekly site visits grew from **2–5 to 80–90**, generating **5 qualified leads per week**.
- Led the team's **AI adoption**: AI-assisted development, customer acquisition and retention strategies, and document analysis.

Instituto Politécnico Nacional (IPN) — ESIME Culhuacán | Software Developer → Cloud Architect & Lead Developer

Mexico City · Mar 2012 – 2026 · Project-based government technology work under IPN collaboration agreements, in parallel with ZOGA; started as a developer and grew into team lead and cloud architect

IAPREP — AI for electoral tally sheets · 2020 – 2025

- Cloud architect** of the team's own AI product: two **PyTorch** models (handwritten text and digits) trained on **16M samples**, with **97% reliability**; serverless AWS API at 0.7 s per region and peaks of **1,000 records/second**. Stress-tested on historical PREP data and rated **best of 3 solutions in INE's 2024 evaluation**.

COFEPRIS — Clinical trials and online sanitary regulation · Jul 2023 – Dec 2024

- Architect and lead developer** of a data-driven workflow engine that digitized **100+ regulatory procedures**, with shared SSO, e.firma and a decoupled payments engine (AWS AppSync, CloudFront, DocumentDB, DynamoDB, **Terraform**).

Electoral systems audits — INE & IEEM · 2016 – 2023

- Forensic replication of INE's 2015 quick-count error using bots and real-data datasets, pinpointing the responsible code.
- Security audit of INE's **internet voting system** for Mexicans abroad with a **cryptographic integrity verification methodology** (2022); hardware and software audits of **4 electronic voting machine models** across 4 elections (2020–2023).
- Real-time code review model for the preliminary results program (PREP) of IEEM, the State of Mexico electoral institute (2021).

Red Integral Notarial — Colegio de Notarios · Nov 2018 – Jul 2021

- Lead backend engineer and cloud administrator**: serverless AWS backend, iOS/Android apps with e.firma and biometric verification, and a secured VPN for **250 notary offices**.

Manahuia — SECTEI (Mexico City government) · 2025 – 2026

- Real-time gas-leak detection with in-house **IoT** devices streaming to AWS, tested with **5,000 simultaneous devices**.

Earlier projects — SEDENA & TIMEX · 2012 – 2014

- Information-security risk management system for SEDENA (Java, Oracle, on-premise); multi-tenant e-invoicing (CFDI) platform for TIMEX handling peaks of **5,000 invoices/minute** (PHP, MySQL, S3), plus a web POS.

Upperbus (later Rideupp) | Co-founder

Mexico City · 2015 – 2017

Urban mobility startup. Launched as **Upperbus**, shared bus routes for commuters with a common origin and destination, then **pivoted the business model to Rideupp**, a carpooling platform. Product leadership, multidisciplinary team and city operations (NativeScript, Node.js, Firebase, AWS).

SELECTED AI PROJECTS

- **rag-agent** (github.com/MrCalamitus/rag-agent, 2026): auditable RAG agent that cites the evidence behind every answer; multi-topic service on **AWS Bedrock** (Knowledge Bases, S3 Vectors, Guardrails for PII) with ECS Fargate, PrivateLink, CloudWatch/X-Ray and **Terraform**. Python/FastAPI, hexagonal architecture, Open Responses API and an **evaluation suite** with reference questions.
- **Hermes-based personal assistant** running on local LLMs: daily research digests, message summaries and expense tracking (Qwen3-VL for receipt OCR with a cloud-model fallback).

SPEAKING

- Workshop "**Security practices in microservices development on AWS**" — 9th Information Security Congress, IPN ESIME Culhuacán (Oct 2021).
- Workshop on **AI agents applied to software development** — IPN (upcoming, Oct 16, 2026).

TECHNICAL SKILLS

Cloud & Infrastructure: AWS (Lambda, AppSync, CloudFront, DynamoDB, DocumentDB, ECS Fargate, Bedrock, S3, RDS, CloudWatch, X-Ray), Serverless, Terraform, Docker, Google Cloud / Firebase, Linux

Development: Node.js, Python, Java, GraphQL, REST APIs, MySQL, MongoDB, Oracle, NativeScript

Artificial Intelligence: Amazon Bedrock (Knowledge Bases, Guardrails), RAG, vector search (S3 Vectors), LLM evaluation, multi-agent systems (CrewAI), local LLMs (Ollama, Qwen), Anthropic API, PyTorch, computer vision / OCR, prompt engineering, AI-assisted development (Claude Code, Cursor)

Security: Security by design, electronic signature and applied cryptography, security audits, regulatory compliance

Leadership: Technical leadership (teams of up to 5), Agile / Scrum, mentoring

EDUCATION

M.Eng. in Information Security and Information Technologies — Instituto Politécnico Nacional (IPN), ESIME Culhuacán · 2020 – 2024

B.Eng. in Computer Engineering — Instituto Politécnico Nacional (IPN), ESIME Culhuacán · 2008 – 2014

LANGUAGES

Spanish — Native · **English** — Conversational / technical